

Network Slicing Using Artificial Intelligence

PG Seminar Presentation

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Motivation & Problem Statement

5G/6G Network Challenges:

- Explosive growth of heterogeneous services
- Diverse QoS requirements:
 - **eMBB**: High throughput (> 1 Gbps)
 - **URLLC**: Ultra-low latency (< 1 ms)
 - **mMTC**: Massive device connectivity
- Static resource allocation is **inefficient**
- Traditional: fail under **dynamic traffic**

Core Problem

How to **dynamically** and **intelligently** allocate limited Physical Resource Blocks (PRBs) across multiple network slices while satisfying slice-specific SLAs?

Key Limitations of Existing Approaches:

- Model-free RL: poor sample efficiency
- Static ML models: no real-time adaptation
- Rule-based: cannot generalize
- LLMs alone: lack low-level control

Research Objective

Design an AI-driven framework combining **RL + LLM** for adaptive and efficient O-RAN network slicing

O-RAN Architecture & Network Slicing

O-RAN Disaggregated Components:

- **RU** – Radio Unit (physical layer)
- **DU** – Distributed Unit (low-layer RAN)
- **CU** – Centralized Unit (high-layer RAN)
- **Near-RT RIC** – Near Real-Time RAN Intelligent Controller
- **xApp / rApp** – Intelligent control applications

Network Slicing Concept:

- Virtualized, isolated logical networks over shared physical infrastructure
- Each slice: independent SLA, scheduler, QoS policy
- Resource Block (RB) allocation: smallest time-frequency unit

O-RAN Architecture & Network Slicing

Three Service Slices

Slice 1 (eMBB)	High throughput, video streaming, cloud gaming
Slice 2 (mMTC)	Massive IoT, device connectivity, smart grids
Slice 3 (URLLC)	Autonomous driving, industrial control, low latency

Control Hierarchy:

- *Inter-slice*: RIC allocates PRBs between slices
- *Intra-slice*: DU schedules PRBs per UE within slice

Task 1: RL for Dynamic Resource Allocation

MDP Formulation:

- **State** $s_t = \{Q_l, N_{l,u}, a_{t-1}\}$: QoS levels, active UEs, previous allocation
- **Action** $a_t = \{b, e\}$: inter-slice PRB partitioning + intra-slice scheduling
- **Reward** $r_t = r_Q - r_{ng}$: QoS-based reward with penalty for violations

Optimization Objective:

$$\max_{b,e} \sum_{l \in \mathcal{L}} w_l U_l(b, e) \quad \text{s.t. SLA constraints}$$

Slice-Specific QoS Metrics:

- eMBB: $\mu_r = \frac{1}{N_u} \sum_i C_i$ (avg. throughput)
- mMTC: $d_s = \frac{\sum \mathbf{1}(C_i > \lambda_i)}{N_u} \sum_i C_i$
- URLLC: $l_d = \max(\tau_i)$ (worst-case latency)

RL Algorithm: Soft Actor-Critic (SAC)

- Continuous action space support
- Entropy regularization for exploration
- MARL: distributed DU actors + centralized RIC critic

Kernel-Based RL (KBRL) – Model-Based Approach

Why Model-Based RL?

- Model-Free requires thousands of steps to converge
- Online learning on live network demands **sample efficiency**
- KBRL learns SLA violation predictor per slice

KBRL vs. Model-Free RL:

- ✓ Convergence in < 5000 steps vs. $>$ for DRL
- ✓ Near-zero SLA violations during learning
- ✓ Lower computational overhead
- ✓ Consistent across all scenarios

CMDP Formulation:

$$\min_{\pi \in \Pi} \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}_{\pi} \left[\sum_{n=1}^N \sum_i A_n^{(i)} \right]$$
$$\text{s.t.} \quad \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}_{\pi} \left[\sum_{n=1}^N I^{(i)} \right] \leq \delta$$

Task 2: LLM Integration – xApp & rApp

Why LLMs for Network Slicing?

- LLMs encode semantic domain knowledge
- Convert raw RF/QoS metrics into **structured context-aware prompts**
- Enable **intent-driven** resource allocation

LLM-xApp Framework (O-RAN Near-RT RIC):

- Monitors UE QoS state via E2 interface
- LLM agent generates PRB allocation decisions
- Uses **OPRO** (Optimization by Prompting)

LLM-rApp Framework (Non-RT RIC):

- Long-term policy management
- RAG-based context retrieval from O-RAN documentation
- Generates slicing policies for xApps

Meta-Prompt Structure

- 1 **Objective:** Maximize QoS across slices
- 2 **History:** Previous allocations + evaluation scores
- 3 **Instructions:** Output (PRB allocation vector)

ORAN: LLM Augmented MARL Framework

Framework Architecture:

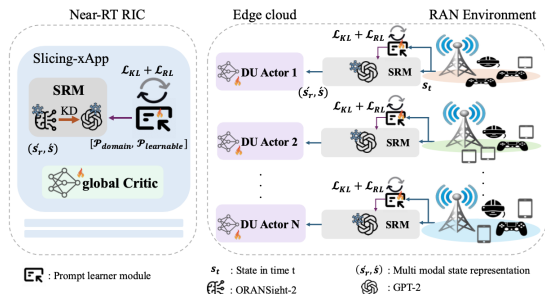


Fig. 1: Overview of the ORAN-GUIDE system architecture showing distributed DRL agents coordinated by a centralized critic, with prompt-augmented State Representation Modules (SRMs).

LLM Xapp Architecture:

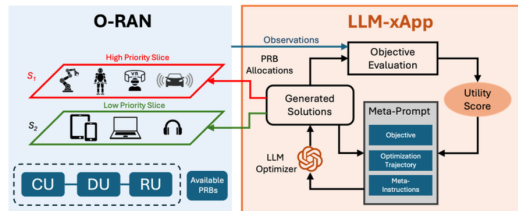


Fig. 2: LLM-driven optimization of resource provisioning in O-RAN.

5G Network Architecture – Implementation

Hardware Setup:

- **SDR Hardware:** $2 \times$ USRP B210
 - One as **gNodeB** (base station)
 - One as **UE** (user equipment)
- **Software:** OpenAirInterface (OAI) 5G stack
- **Core Network:** OAI 5G CN via Docker containers:
 - oai-amf, oai-smf, oai-upf, MySQL database
- **Bandwidth:** 20 MHz, 15 kHz subcarrier spacing



Fig. 3: Two USRP B210 SDRs: gNodeB (green LED) and UE (red LED)

Achieved

Full end-to-end 5G NR link: UE $\leftrightarrow$ gNB $\leftrightarrow$ Core Network – **NAS Registration + PDU Session Established**

5G Simulation & Network Parameter Analysis

Observed PHY Layer Parameters:

- SSB Detection, Cell ID confirmed
- Sync source peak: **59 dB** (strong signal)
- Frame/Slot tracking: Slots 128, 256, 384 ..
- PBCH decoding: CellId detected, TDD Normal prefix

Core Network Status (Docker):

- All CN components: **Up 4+ hours (healthy)**
- Ports: 80/tcp, 5342–5344/tcp, 8080/tcp
- NAS Messages:** Registration Accept + PDU Session Establishment Accept

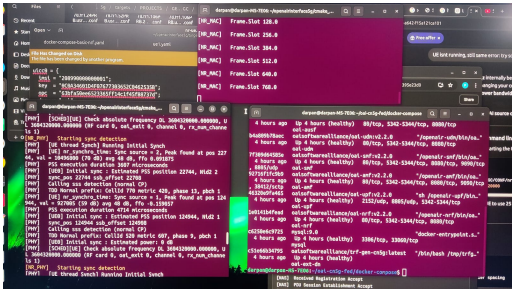


Fig. 4: OAI simulation: PHY-layer sync logs (left) and 5GCN Docker containers (right)

USRP Configuration

DL Freq: 3604.32 MHz — UL: 3604.32 MHz
RF card 0, oai_exit 0, rx_num_channels 1

Thesis Overview & Work Plan

Phase 1 – Completed:

- ✓ Full 5G stack setup (OAI + USRP B210)
- ✓ End-to-end NR link: UE ↔ gNB ↔ 5GCN
- ✓ Literature Review: for network slicing
- ✓ Problem formulation: MDP + CMDP modeling
- ✓ RL framework design (SAC-MARL + KBRL)

Phase 2 – Ongoing:

- RL agent implementation & simulation
- O-RAN slicing simulator (eMBB/URLLC/mMTC)
- Performance benchmarking vs. baselines

System Architecture Summary

Non-RT RIC (rApp + LLM)



Near-RT RIC
xApp (RL Agent + LLM-xApp)



Distributed Units (DUs)
SAC Actors — PRB Allocation



RAN Environment
eMBB, URLLC, mMTC Slices

Evaluation Metrics: QoS satisfaction rate, PRB efficiency, SLA violation rate, convergence speed, per-slice CDF throughput

Conclusion & Expected Contributions

Summary:

- Proposed a two-task AI-driven O-RAN slicing framework:
 - ① **RL (SAC-MARL / KBRL)**: Dynamic PRB allocation satisfying heterogeneous slice SLAs with high sample efficiency
 - ② **LLM Integration (xApp + ORAN-GUIDE)**: Semantic state representation and intent-driven resource allocation using dual-LLM + RAG
- Full 5G network (OAI + USRP B210) operational
- Demonstrated end-to-end NR link with 5G core

Novelty to Introduce:

- Causal understanding of network dynamics (Causal reasoning + constraint-aware prompting)

Expected Outcomes

- > 14% relative performance improvement over baseline MARL
- Near-zero SLA violations during online learning
- Scalable, edge-deployable AI slicing controller

References

-  F. Lotfi, H. Rajoli, and F. Afghah, “ORAN-GUIDE: RAG-Driven Prompt Learning for LLM-Augmented Reinforcement Learning in O-RAN Network Slicing,” *arXiv preprint arXiv:2506.00576*, 2025. 0
-  J. J. Alcaraz, F. Losilla, A. Zanella, and M. Zorzi, “Model-Based Reinforcement Learning with Kernels for Resource Allocation in RAN Slices,” *IEEE Transactions on Communications*, 2021. 1
-  X. Wu, J. Farooq, Y. Wang, and J. Chen, “LLM-xApp: A Large Language Model Empowered Radio Resource Management xApp for 5G O-RAN,” *Proc. NDSS FutureG Workshop*, 2025. 2
-  B. Liu, J. Tong, and J. Zhang, “LLM-Slice: Dedicated Wireless Network Slicing for Large Language Models,” *ACM SenSys*, 2024. 3
-  T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor,” *ICML*, 2018.
-  O-RAN Alliance, “O-RAN Architecture Description,” White Paper, 2022.

Thank You

Questions & Discussion

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